Department of Data Science and Artificial Intelligence
National Institute of Technology Kurukshetra
B.Tech (AI&DS), **DSPC 201/Foundations of Data Sc.**, Mid Term (I) Exam, ACY. 2025-26

Time Allowed: 50 Mins
Max. Marks: 15.00

Date of Exam: 26.09.2025
Roll No:_ _ _ _ _ _ _ _ _

Note: All the questions and their parts are compulsory; there is no choice. Answer each part of a question in the same place.

Q1. Anwar the following **briefly**: (1*4)

(a). How would you define *Data Science*? Support your answer with a real-world case that emphasizes its practical relevance.

(b). Briefly write concise notes on each, explaining their *conceptual definitions* and *task*: *Data Modeling, Data Retrieval, Data Visulaization, Data Anlysis & Data mining.*

(c). Which graph is effective for outlier detection, between *box-plot* and *scatter plot?* Why?

(d). Why feature selection is considered as *NP problems*, rather than *NP-hard*, justify your answer with a suitable example.

Q2. (a) How notions of '*Correlation* and '*Covariance*' play important role in data reduction? Prove whether two data objects *A(4,2,2,2,7,3,6)* and *B(4,6,3,7,1,2,3)* are correlated or not ? (2.00)

(b) A hospital tested the *age* and body *fat(%)* data for 20 randomly selected adults with the following results: (3.00)

age	23	23	27	27	41	41	47	49	50	50
%fat	9.5	26.5	37.8	17.8	45.5	25.9	47.4	27.2	31.2	42.5

age	52	54	54	54	54	58	58	60	61	64
%fat	34.6	42:5	28.8	33.4	39.0	54.1	32.9	41.2	35.7	40.1

Calculate the *mean, median,* and *standard deviation* of *age* data. Estimate the mode of the *age* data? Comment on the *age* data's modality (*i.e., bimodal, trimodal, etc.*). Draw the *Box-plots* for *%fat*.

Q3. (a). Suppose you want to study the eating habits of students at *NIT Kurukshetra*. Students come from different states across India and have diverse cultural backgrounds, dietary restrictions, and cuisine preferences. Additionally, some students live in hostels with mess facilities, while others prefer cooking their own meals or ordering food from outside. Your study aims to understand patterns in meal frequency, food choices, and cultural influences on diet. Which sampling technique would you adopt? Justify your choice and discuss any possible selection bias and sampling errors in your method. (2.00) (4.00)

(b). *Is gender independent of education level*? A random sample of 395 people was surveyed and each person was asked to report the highest education level they obtained. The data that resulted from the survey are summarized in the following table. Are gender and education level dependent at a 5% (0.05) level of significance? Perform a *Chi-Square* test, to verify the hypothesis.

7.815

	High School	*Bachelors*	*Masters*	*PhD*	**Total**
Female	60	54	46	41	201
Male	40	44	53	57	194
Total	100	98	99	98	395